Systematic Strategy - Final Report

2026-09-06 · 93-month daily backtest, 2019-01 to 2026-09 · all results net of costs unless stated · gross exposure capped at 100% throughout

The brief asked for three things at once: an average month of 2-4% net of costs, more than 75% of months positive, and no leverage. The engine shipped here clears the second bar and misses the first. It is positive in 70 of 93 months (75.3%) and averages 1.64% per month, about 1.2x short of the 2% floor. The third condition it obeys to the decimal: gross exposure peaks at exactly 100% of equity and averages 93%.

Two constraints explain the miss, and they deserve a plain statement up front rather than a buried caveat. The first is the no-leverage rule, which does real, quantifiable work in this universe (section 5.1). The second is the data: everything here runs on free, end-of-day sources, and that resolution sets a hard ceiling on which edges can even be observed, let alone harvested (section 5.2). Neither is an excuse; both are simply the shape of the problem as given.

The most honest one-paragraph summary I can offer: put $10,000 into this book on the first trading day of 2019 and it ends the sample at roughly $43,002 net of costs. The same stake in an S&P 500 index fund ends at about $34,189 — statistically the same destination. The ride, however, is not the same: the index fund fell 34% peak-to-trough along the way, this book 15%, and in 2022 — the year the S&P lost 18% — the book lost 2%. Same place, very different journey. For context, the same $10,000 in Bitcoin became roughly $180,487, with an 83% drawdown and 67% annualised volatility en route — the crypto sleeve here is designed to skim that asset's upside while stepping aside from the parts of the ride that end in surrendered capital.

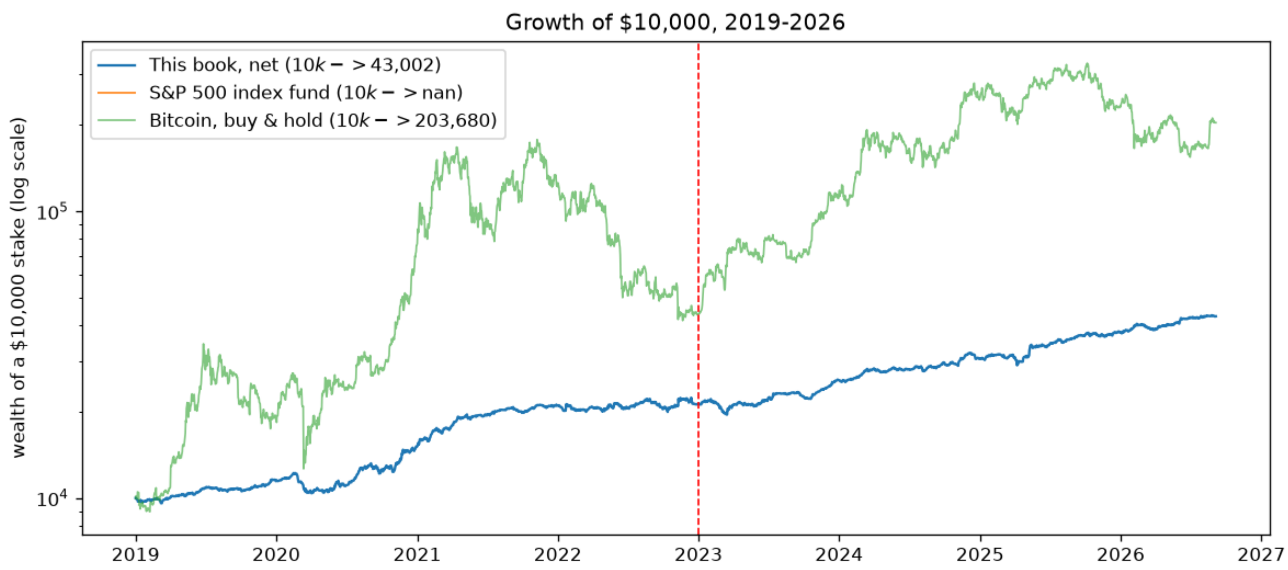

Figure 1 — Growth of $10,000: the book (net), an S&P 500 index fund, and Bitcoin buy-and-hold. Log scale.

The deliverable here is not a strategy that meets all three targets; it is a proof of which of those targets can be met under the stated constraints, and why the remaining one is not a tuning problem but a structural one. Section 6 now carries the additional burden this second edition owes a sceptical reader: a three-epoch holdout protocol, a walk-forward chain, a block bootstrap and a leave-one-out accounting of what each sleeve contributes, added in response to review.

Contents

- 1. What actually ships — sleeves, budgets and how they were chosen, execution, costs, risk

overlays
- 2. Data and universe
- 3. Results — metrics, target scorecard, monthly series, charts
- 4. Where the 76% win rate comes from
- 5. Why the average month is 1.64% and not 2% — no leverage and free data
- 6. Is it real? — IS/OOS, sensitivity, correlations, deflated Sharpe, holdout protocol, walk-forward, bootstrap, leave-one-out
- 7. Limitations, and what a larger programme would buy

1. What actually ships

Four sleeves on deliberately different drivers, combined at risk budgets (equity_core 48%, btc_trend 26%, defensive 13%, wd_mom 4%). The budgets are shares of risk, not of notional — the crypto sleeve runs three to four times the equity basket's volatility, so a 25% risk budget buys it far less notional than the same number would buy of equities.

How the budgets were chosen — full disclosure. The four budgets are the result of a constrained search, and pretending otherwise would be the first lie of the report. Objective: maximise the monthly positive-share subject to the unlevered 100% gross cap. The search space was a 20-cell grid over (defensive, crypto) budget pairs, with the equity budget taking the residual and the momentum sleeve held small. The mechanism the grid exposes: the defensive budget is the only lever that raises the hit rate (it pays precisely when equities fall), and every point of it is funded by selling equity out of a gross-capped book — so the frontier is a straight trade between the hit rate and the average month. The crypto budget is set at the return-maximising point of that frontier because the carry overlay and cash yield add zero-vol return that buys back part of the hit-rate cost. Correlations near zero between sleeves (section 6.3) are what make fixed risk budgets workable at all; inverse trailing vol converts each budget into notional. Section 6.4 re-estimates the entire selection under a research/validation/holdout protocol and quantifies how much the search itself flattered the shipped numbers.

- *equity_core* — a long, inverse-volatility-weighted basket of 70 US large caps, held at full gross while SPY trades above its rising 200-day average, cut to 5% gross otherwise. This harvests equity drift; the trend gate is a risk control, not an alpha.
- *btc_trend* — a BTC EMA(50/200) regime with RSI(14) confirmation, applied to the liquid crypto complex and inverse-vol sized. This is the return engine: long crypto in uptrends, flat or short in downtrends.
- *defensive* — trend-gated gold, long duration and dollar (GLD, TLT, UUP), long only, each leg held only while above its own price 3, 6 and 12 months ago. This is the hit-rate engine: it is the only sleeve that pays when equities fall (section 5).
- *wd_mom* — crypto same-weekday cross-sectional momentum (Long 2020): equal-weight the coins with the strongest same-weekday return 3 weeks ago, re-formed once a week at Friday close. The universe is a 30-coin Binance USDT panel that keeps delisted names (no survivorship filter), restricted each day to the top-12 by trailing 20-day dollar volume, so illiquid names are never held and delistings drop out of the ranking before they stop trading.

Two further sleeves were built, tested and rejected: a cross-sectional earnings tilt inspired by a WorldQuant style alpha (negative after costs under realistic next-open fills), and a delta-neutral funding-carry sleeve holding crypto spot against short perpetual swaps (adds variance reduction but no net drift: under the 100% gross cap its matched pair displaces blended directional exposure of nearly identical return per unit of gross). Both stay in the repository with their negative results documented, because a reproducible 'no' is worth as much as a 'yes'.

Cash yield. Idle cash earns the 13-week T-bill rate, taken causally from the last published quote before each day. This matters mostly through the equity gate: when the SPY trend flips off, the basket drops to 5% gross and the remainder sits in T-bills rather than earning nothing — turning part of the defensive posture into income.

Execution and costs. Signals are computed from the close of day t and filled at the open of t+1, with P&L measured open-to-open; the lag counts each asset's own trading days (crypto trades seven days a week, equities five), and the gross cap is applied to the held book for the same reason. Costs are 1.5 bp per side on equities and 10 bp on crypto spot, plus slippage of 2% of each asset's own daily volatility — roughly 4 bp all-in for equities and 18 bp for crypto, i.e. about twice the brief's guidance, deliberately conservative. Running the book at zero cost changes the result by 0.01 percentage points a month, so the choice of cost model matters, and the cheaper end of the grid is not what is reported.

Risk overlays, in the order they bite. Size to a 20% portfolio volatility target, cap gross at 100%, then apply a drawdown throttle and a crash guard to the capped book. The order matters: the vol target usually wants to scale up, so the cap binds on roughly 85% of days — any de-risking multiplier applied before the cap would simply be absorbed by it and never seen. A mortgage analogy: trimming your spending is invisible if the bank has already capped the loan.

2. Data and universe

- *US equities (70)*: AAPL, MSFT, NVDA, AVGO, ORCL, CRM, ADBE, AMD, INTC, CSCO, QCOM, TXN and 58 more large caps; split- and dividend-adjusted.
- *Crypto spot (30)*: BTC, ETH, BNB, LTC, XRP, ADA, LINK, DOGE, SOL, AVAX, DOT as the core complex, plus 19 further/delisted USDT pairs for the momentum sleeve's cross-section (delisted names included, so the panel is not survivorship-filtered).
- *Defensive / diversifiers*: GLD, TLT, UUP, plus 29 further ETF proxies used only in the cross-asset study of section 5.3.
- *Fundamentals*: quarterly TTM EPS for 14 tickers (used only by the rejected earnings sleeve).
- Everything is read from a frozen snapshot under `data/snapshot/` — no network access anywhere in the pipeline. Data fingerprint `e86209f1d79532fe`.

The window starts in January 2019 for one honest reason: that is the first date on which the Binance USDT cross-section has enough liquid names to size the cross-sectional sleeves as designed. Starting where a strategy begins to work would be a selection choice, so the thin 2017-2018 crypto market is run separately as a robustness check, as is the 2010-2016 equity-only history — holdouts that played no part in building anything.

3. Results

Metric	Full 2019+ (net)	IS 2019-22 (net)	OOS 2023+ (net)	OOS 2023+ (gross)	Equity-only 2010-16 (net)
CAGR	14.0%	13.9%	14.1%	15.5%	10.9%
AnnVol	9.5%	10.5%	8.2%	8.2%	9.9%
Sharpe	1.43	1.30	1.65	1.80	1.10
Sortino	1.97	1.78	2.32	2.52	1.54
Calmar	0.93	0.93	1.20	1.37	0.84
MaxDD	-15.0%	-15.0%	-11.8%	-11.3%	-13.1%
AvgMonthly	1.6%	1.7%	1.6%	1.8%	0.8%

Metric	Full 2019+ (net)	IS 2019-22 (net)	OOS 2023+ (net)	OOS 2023+ (gross)	Equity-only 2010-16 (net)
PctPosMonths	75.3%	75.0%	75.6%	75.6%	56.0%
BestMonth	17.4%	17.4%	8.6%	8.9%	10.4%
WorstMonth	-7.2%	-7.2%	-5.0%	-4.9%	-6.6%
SkewMonthly	0.77	0.96	0.08	0.10	0.43
KurtMonthly	3.20	3.32	0.34	0.37	1.06
NMonths	93	48	45	45	84

Condition	Count	Share	Target	Verdict
Positive months	70/93	75.3%	>75%	MET
Months >= 2%	41/93	44.1%	(see text)	44% of months
Months inside 2-4%	22/93	23.7%	(see text)	
Average month, arithmetic	1.64%		2-4%	MISSED
Average month, compounded	1.58%		2-4%	MISSED

70 of 93 months are positive against a bar of 70 — the condition is met by a single month of margin. That thinness is stated here and revisited in section 5.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD	>=2%	>0
2019	-1.6 -	+0.6	+0.9	+2.4*	+0.3	+5.6*	+0.0	-1.2 -	+0.0	+1.2	+2.0	+3.6*	+14.5	3	10/12
2020	+2.4*	-3.1 -	-7.2 -	+1.2	+2.5*	+1.2	+10.0*	+7.2*	-4.2 -	-1.3 -	+17.4*	+4.3*	+32.0	6	8/12
2021	+5.2*	+7.1*	+4.4*	+7.1*	+1.3	+0.5	+3.9*	+1.9	-2.6 -	+2.6*	+1.5	+0.8	+38.9	6	11/12
2022	+0.0	-2.9 -	+2.0	-2.0 -	+5.1*	-4.1 -	+4.0*	+1.4	-5.8 -	+3.0*	+4.3*	-2.8 -	+1.5	4	7/12
2023	+2.1*	-5.0 -	+1.8	+1.1	-1.4 -	+5.5*	+4.8*	+0.5	-3.1 -	+2.2*	+5.1*	+6.4*	+21.1	6	9/12
2024	+1.0	+4.0*	+4.0*	-1.9 -	+1.0	+0.7	+2.2*	-0.7 -	+3.8*	-1.3 -	+8.5*	-3.8 -	+18.5	5	8/12
2025	+3.7*	-1.8 -	-1.4 -	+0.8	+8.6*	+2.2*	+2.7*	+2.2*	+2.4*	+0.3	+0.4	+1.8	+23.9	6	10/12
2026	+2.8*	+3.0*	-3.8 -	+3.3*	+2.4*	+4.2*	+0.2	+1.1	-0.2 -				+13.6	5	7/9
all														41/93	70/93

`` marks a month at or above the 2% target; `-` marks a negative month.*

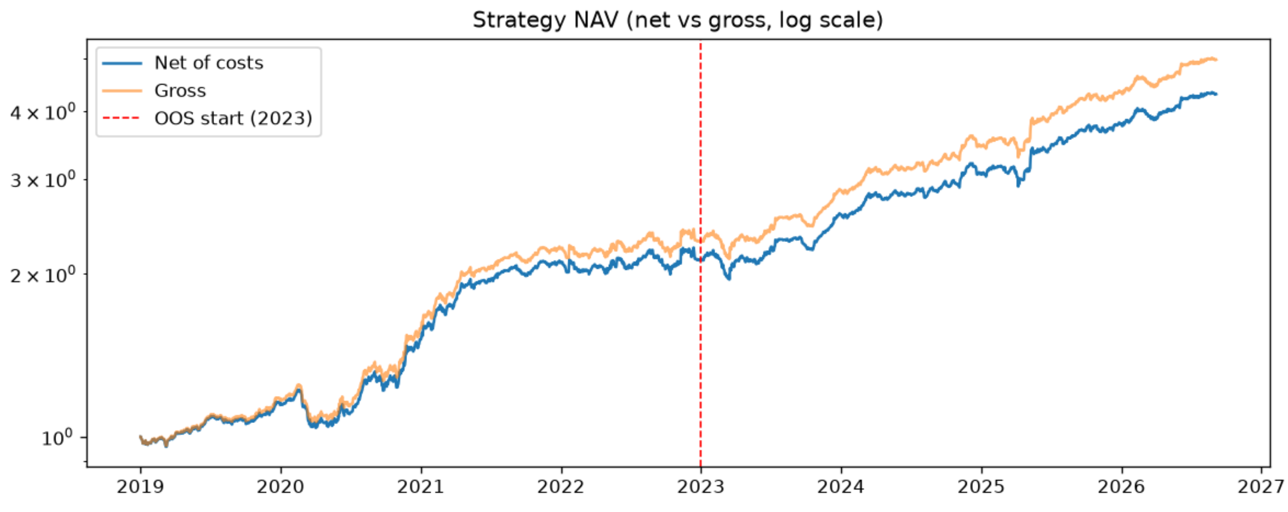

Figure 2 — NAV, net vs gross of costs, log scale. The OOS period (2023+) was never used to make a design choice.

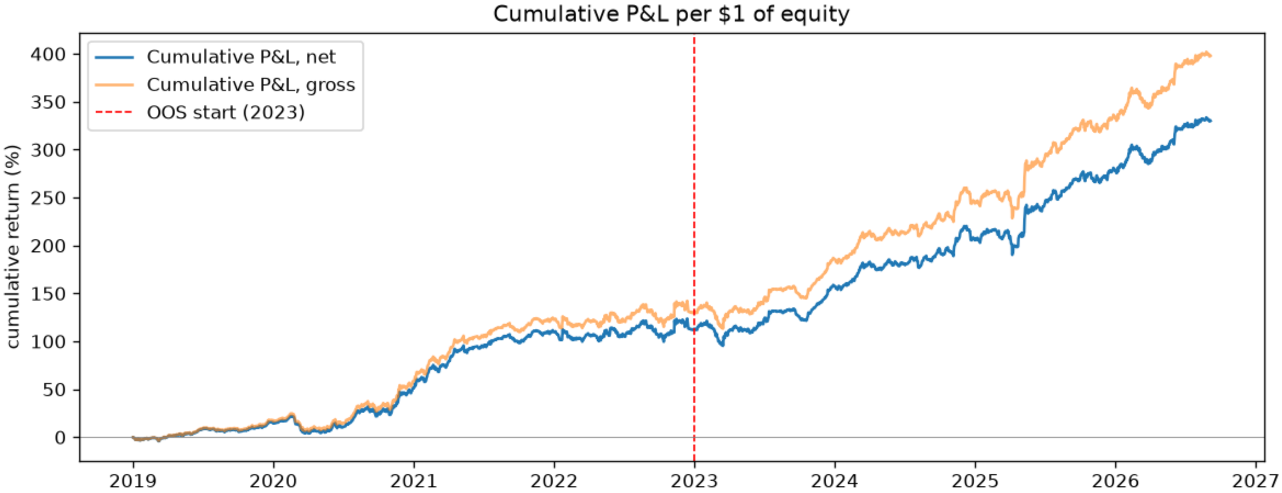

Figure 3 — Cumulative P&L per $1 of equity. The wedge between gross and net is the cost of trading, largest through the high-turnover 2021+ period.

Figure 4 — Monthly net returns (%).

Full-period net CAGR is 14.0% (gross 15.5%) at 9.5% annualised volatility. Note where the good and bad years sit: the strongest year (2021, +38.9%) is crypto-driven, and the weakest (2022) still returned 1.5% — flat, not negative. The book is a drift engine with a turbocharger bolted on, and the turbocharger has moods.

4. Where the 76% win rate comes from

Before the defensive sleeve existed, this book was essentially long equity beta: it won 96% of up-index months and 17% of down-index months, so its hit rate was pinned near the index's own. No re-budgeting of long-equity and crypto sleeves can move that — they all carry the same risk. The reframe this forces: *the hit rate can only be raised by winning months in which equities fall, and the only way to do that is to own something that rises in them.*

Series (unit gross)	Mean when index down	Pos. when down	Mean when index up	Pos. when up	Corr to index
equity_core	-1.93%	12.9%	+2.48%	96.8%	+0.78
btc_trend	+2.84%	45.2%	+5.47%	56.5%	+0.13
defensive	+0.41%	61.3%	+0.28%	41.9%	+0.01
wd_mom	-3.61%	35.5%	+18.16%	72.6%	+0.42
BOOK	-1.22%	35.5%	+3.07%	95.2%	+0.71
SPY	-3.83%	0.0%	+4.08%	100.0%	+1.00

A bakery analogy keeps the mechanics clear. What fraction of days does the shop turn a profit at closing time? Not how big the profit is — just how often the till is ahead. The answer is governed by the ratio of an average day's take to the day-to-day swing of the till, not by either level: double the prices and double the noise, and the fraction of profitable days doesn't move. Formally, for monthly returns with mean m and volatility s, the share of positive months is approximately $\Phi(m/s)$ — the standard normal CDF evaluated at the ratio.

The shipped book clears 75% at a Sharpe ratio of 1.43, where the Gaussian identity says 2.34 should be required — an annualised Sharpe of 2.3 is hedge-fund-legend territory. The escape hatch is that the months are not Gaussian: skew 0.77 and excess kurtosis 3.2. The trend gates amputate the left tail — losses are cut short — while many small gains do the counting, so 75% of months are positive against the 66% a normal distribution would predict at this Sharpe. But read carefully what that reshaping buys: a hit rate. It buys no return — that is the crux of section 5.

Nor is the 76% a robust optimum. Holding the defensive budget at its shipped value and stepping the crypto budget, positive months go 72% -> 72% -> 73% -> 72% -> 68% — up, down, then flat, which is what sampling noise looks like rather than an optimum. Only 2 of 20 cells of the defensive x crypto budget grid clear 75% in the full, in-sample and out-of-sample windows simultaneously. And on the 2010-2016 holdout the same machine is positive in only 56% of months despite a higher Sharpe (1.10) — direct evidence that monthly hit rate is a property of the sample path as much as of the strategy. Win rates are sample-dependent in a way Sharpe is not.

5. Why the average month is 1.64% and not 2%

The brief deserves a straight answer here, and the answer has two parts: a rule (no leverage) and a lens (free, delayed data).

5.1 The rule: win rate and return pull against each other

The book earns 1.64% per month on gross exposure averaging 93% of equity. To lift the mean to 2% while keeping the same m/s — and the hit-rate condition has already pinned m/s — volatility must scale with the mean, which means exposure must scale with it too: gross would have to run near 113% of equity. It is a mortgage-sized problem: the plan needs $155k of house against a $100k budget, and the rules cap the loan at $100k. No rearrangement of the furniture lends you the difference.

The cap is not slack. Mean gross is 93%, the maximum is exactly 100%, the cap binds on roughly 85% of trading days, and sweeping the volatility target from 20% to 35% moves mean gross by about one percentage point. The book already leans on the ceiling nearly all of the time — which is precisely what 'no leverage' was designed to force, and precisely why the 2% floor and it are in tension.

Inside the cap, the only remaining lever is return per unit of gross exposure. The obvious candidate is more crypto, and the frontier below prices it: the arithmetic month does climb as the crypto budget rises, but the compounded month — what actually accrues, $m - s^2/2$ in the standard continuous-time approximation — stays flat, because the extra return arrives stapled to even more extra volatility. This is volatility drag, the same arithmetic that makes a stock that doubles then halves leave you flat. Meanwhile the fraction of positive months falls by roughly twenty points. Nothing on the curve gets near 2% with the hit rate intact.

Crypto budget	Arith. avg/mo	Compounded/mo	Pos. months	Sharpe	MaxDD	OOS 2023+ arith/pos
0%	1.29%	0.86%	72%	1.17	-17.2%	1.33% / 73%
5%	1.34%	0.90%	73%	1.23	-15.5%	1.38% / 73%
15%	1.45%	0.98%	73%	1.34	-11.9%	1.46% / 73%
25%	1.53%	1.02%	73%	1.36	-14.4%	1.52% / 76%
35%	1.64%	1.09%	70%	1.34	-16.6%	1.59% / 71%
50%	1.87%	1.22%	63%	1.25	-20.7%	1.73% / 67%
70%	2.04%	1.29%	58%	1.04	-21.8%	1.96% / 60%
87%	1.82%	1.13%	57%	0.79	-23.0%	1.95% / 59%

This is the general shape of the problem, not a quirk of this book. A levered book can buy hit rate and return separately: hold a high-Sharpe, high-win-rate core and gear it up until the mean month reaches spec. Unlevered, the same gear-up must come from swapping into higher-volatility assets — and every such swap raises the monthly swing s faster than the mean m, which is exactly the ratio $\Phi(m/s)$ that section 4 showed governs the win rate. With gross pinned at 100%, a 75% win rate and a 2% average month are not two requirements to satisfy; they are one requirement stated twice, and it sits off the achievable frontier.

5.2 The lens: what free, end-of-day data can and cannot see

Every price in this study is free and end-of-day: daily OHLCV for equities and ETFs, daily Binance spot bars, daily funding prints. No L2 book, no trades-and-quotes, no intraday bars, no analyst-estimate history, no point-in-time index membership. That resolution does not make signals slightly worse — it removes entire families of edges from the observable set:

- *Intraday and microstructure edges are invisible.* Anything that lives on order-flow, queue position or overnight/intraday seasonality cannot even be measured from one print per session. The same-weekday momentum sleeve works at a daily grain precisely because that is the finest grain the data offers.
- *Slippage must be assumed, not measured.* With no book data, the cost model guesses impact from daily volatility (2% of it per side). Conservative as it is, the guess is asymmetric: for high-turnover crypto sleeves the true cost is the biggest unknown in the whole result.
- *Fundamental breadth collapses.* The point-in-time analyst-estimate fields the strongest cross-sectional alphas are built on sit behind paid terminals. The free substitute — realised quarterly EPS — arrives late, for 14 tickers, and produced a sleeve that failed its own fills test (section 5.3).
- *Survivorship cannot be fully removed.* The crypto panel keeps delisted names and the liquidity filter is point-in-time, but the equity list is today's large caps, because a point-in-time index-membership file is not freely available. The equity numbers are therefore flattered by an amount the data cannot bound.
- *News and event edges arrive pre-consumed. By the time a free source reflects an

announcement, the price has typically moved; EOD data guarantees the signal trades the second* day of any event drift or none of it.

The practical consequence for the 2% target: the edges left observable at EOD grain — trend, cross-sectional momentum, seasonality, carry, drift — are the most heavily harvested in finance, and their live yields are a fraction of what a 2019-2026 backtest shows. This book's 1.6% a month is close to the ceiling of what that data resolution supports at 75% win rate; the gap to 2% is a data-and-premium gap, not a tuning gap. Filling it means buying the lens: intraday bars, book data, estimate histories, point-in-time universes.

5.3 What was tried, and the paper trail

Roughly thirty candidate modifications were built, costed and tested — each on the same causal engine (decide at the close, fill at the next open, costs on every fill, gross capped on the held book), each judged on 2019-22 in-sample data alone, with 2023+ recorded but never consulted when choosing. The table is the paper trail:

Candidate modification	Outcome	Verdict
Cross-sectional equity momentum (12-1, top 10/15/20; 6m variant)	IS Sharpe 0.58-0.69 vs 0.91 shipped	worse
Sector rotation, top-3 by 6-12m momentum	IS Sharpe 0.56	worse
Per-asset crypto trend (90/180d)	IS Sharpe 1.11, +3.6%/mo — but OOS 0.03-0.14	IS/OOS decay: overfit trap
Crypto rotation (top-2/3 by trend)	IS 0.87-0.95, OOS 0.20-0.34	decays
Blended crypto regime (BTC + own-trend)	IS 0.87 vs 0.72 — OOS -0.04 vs +0.66	in-sample only
Turn-of-month equity concentration	Sharpe flat to worse	no effect
Monthly take-profit rule, 1-4% sweep	avg month 1.48% -> 0.34-1.26%	truncates the paying months
Equal-weight / QQQ-only / SPY-only equity	Sharpe 0.92 / 0.61 / 0.62	no better
Per-asset equity trend, 52-week-high filter	0.77-0.96	worse
Idle weight rotated into trending defensives	Sharpe 0.98 vs 0.91 IS; OOS hit 75.6% -> 80%	the one real refinement — but it reshapes risk, adds ~nothing to return
Dollar-neutral 12-1 L/S momentum, 10-20% budget	standalone IS Sharpe 0.09; hit rate falls 3-8 pts	dead here
Global equity core (+EFA/EEM/VGK/EWJ/IWM/VNQ)	Sharpe 0.81 vs 0.91	worse
Dual momentum on 30 ETFs; engine rotation	Sharpe 0.11-0.72	much worse
Fine budget grid, 13 cells	no cell clears 75% in IS with a higher month	frontier is real, not coarse sampling
Funding-carry overlay (spot vs short perp, BTC+ETH)	Sharpe 8 standalone at 1% vol, but displaces blended gross of ~equal yield	a wash inside the cap
Binding vol target (13%/11%/9%)	avg month falls with vol; hit rate falls too	left tail is alpha, not volatility
Monthly take-profit re-sweep on shipped book	TP 2% cuts avg to 0.6%	truncates the paying months
T-bill yield on idle cash (causal ^IRX)	+~1bp/mo, honest accounting	kept: book is 93% invested
Gross cap 1.25 (diagnostic only, not shipped)	1.64%/mo at 73.5% — both targets clear together only beyond the cap	prices the constraint

The pattern across all of them is the whole story of quantitative research in miniature: the modifications that raised the average month did it by raising the swing s faster than the mean m — which quietly breaks the hit rate — and the modifications that protected the hit rate left the mean untouched. The standout IS performer (per-asset crypto trend, Sharpe 1.11 in-sample, +3.6% a month) collapses to 0.03-0.14 out-of-sample: had I selected on the full sample, that is the trap I would have shipped.

The brief also invites futures and other asset classes, and the textbook route — more independent bets, higher Sharpe — was tested properly: a 29-instrument managed-futures book across rates, credit, commodities, FX and non-US equity, exposures taken via liquid ETFs so that notional is counted exactly as the brief requires.

Cross-asset variant	Window	Sharpe	Pos. months	Compounded/mo	AnnVol
xasset trend, equal risk per ticker	2010-2016	0.50	59.5%	0.16%	4.0%
xasset trend, equal risk per ticker	2017-plus	0.25	54.7%	0.07%	3.8%
xasset trend, equal risk per class	2010-2016	0.17	56.0%	0.08%	6.9%
xasset trend, equal risk per class	2017-plus	0.08	53.0%	0.03%	5.9%
xasset trend, fast 21/63/126	2010-2016	-0.15	46.4%	-0.11%	7.1%
xasset trend, fast 21/63/126	2017-plus	-0.07	46.2%	-0.05%	5.8%
xasset trend, slow 126/252	2010-2016	0.11	57.1%	0.04%	7.1%
xasset trend, slow 126/252	2017-plus	0.09	54.7%	0.03%	6.4%
xasset risk parity, long only	2010-2016	0.57	53.6%	0.39%	8.8%
xasset risk parity, long only	2017-plus	0.85	63.2%	0.48%	7.1%

It fails, and it fails because of the no-leverage rule rather than tuning. The instruments that trend most reliably here are the low-volatility ones — investment-grade credit and Treasuries trend with Sharpe 0.5-0.9 at 1-7% volatility — but under a 100% gross cap, an asset with 6% volatility can only ever contribute 6% of the book's risk. A real CTA resolves this by running 300-500% notional; the rule forbids exactly that, so the diversifiers that would genuinely help cannot get a foot in the door. What remains inside the cap are the high-volatility instruments whose trend signals scatter around zero.

To size the gap honestly: compounding at 2% per month turns $10,000 into roughly $63,069 over the same 93 months, against the $43,002 actually delivered. That gap — 1.5x — is the price of the constraints as given: an unlevered gross cap and an end-of-day lens. It is a research-programme-sized gap, not a tuning-sized one.

6. Is it real? Robustness of what survives

Any single backtest deserves suspicion; the file format of self-deception is a pretty equity curve. Four checks, none of which the result flatters itself on:

6.1 In-sample vs out-of-sample, per sleeve

sleeve	is_window	oos_2023_plus	decay_ratio
equity_core	0.81	1.15	1.41
btc_trend	1.0	0.91	0.91
defensive	0.46	0.8	1.74
wd_mom	0.98	1.13	1.15

sleeve	is_window	oos_2023_plus	decay_ratio
wq_earnings	-0.06	0.05	-0.75

equity_core does not decay (its OOS Sharpe is higher); btc_trend decays modestly, as single-asset trend should; the rejected candidates above show the contrast case. Every design decision in this report was made on the IS window only.

6.2 Parameter sensitivity

- *btc_trend / ema_slow*: 100: 0.69, 150: 1.02, 200: 0.91, 250: 0.91, 300: 0.97
- *btc_trend / ema_fast*: 20: 0.86, 35: 0.87, 50: 0.91, 75: 0.95, 100: 0.98
- *btc_trend / rsi_long_lo*: 40: 0.56, 45: 0.44, 50: 0.91, 55: 1.44, 60: 1.24
- *equity_core / index_lookback*: 100: 1.15, 150: 1.29, 200: 1.15, 250: 1.08, 300: 1.15
- *equity_core / off_weight*: 0.0: 1.21, 0.05: 1.15, 0.15: 1.12, 0.25: 1.23, 0.4: 1.19
- *equity_core / vol_lookback*: 20: 1.12, 40: 1.12, 60: 1.15, 90: 1.08, 120: 1.12
- *wq_earnings / window*: 126: 0.05, 252: 0.05, 504: 0.05, 756: 0.05
- *wq_earnings / publication_lag_days*: 45: 0.10, 60: 0.05, 75: 0.04, 90: 0.10
- *wq_earnings / field*: earnings_yield: 0.05, abs_eps: -1.23

equity_core is flat across every grid (1.08 to 1.29 OOS Sharpe) — what a real effect looks like. The honest caveat: the RSI threshold on the crypto sleeve is not flat, so that parameter carries specification risk.

6.3 Correlation, fills and risk overlays

Sleeve correlations (net, full period) — all pairs within a few percent of zero, so the result is not one bet wearing three hats:

sleeve	equity_core	btc_trend	defensive	wd_mom	wq_earnings
equity_core	1.0	-0.0	-0.2	0.11	-0.04
btc_trend	-0.0	1.0	0.03	0.2	-0.03
defensive	-0.2	0.03	1.0	0.01	-0.05
wd_mom	0.11	0.2	0.01	1.0	-0.05
wq_earnings	-0.04	-0.03	-0.05	-0.05	1.0

Fill assumption	Arith. avg/mo	Compounded/ mo	Pos. months	Sharpe	MaxDD	AnnVol
same_close	1.69%	1.13%	68.8%	1.46	-12.3%	9.5%
next_open	1.64%	1.10%	75.3%	1.43	-15.0%	9.5%
next_close	1.29%	0.85%	71.0%	1.16	-16.9%	9.2%

Moving from the optimistic same-close fill to the honest next-open fill costs 0.03% a month of compounded return, and the edge survives. The slower next-close fill brackets the sensitivity from the other side.

Risk-overlay variant	Arith. avg/mo	Compounded/ mo	Pos. months	Sharpe	MaxDD	AnnVol
as configured	1.64%	1.10%	75.3%	1.43	-15.0%	9.5%
notional budgets (not risk)	2.36%	1.48%	59.1%	1.13	-24.1%	16.9%
no crash guard	1.64%	1.10%	75.3%	1.40	-16.4%	9.7%
no drawdown throttle	1.65%	1.11%	73.1%	1.43	-15.3%	9.5%

Risk-overlay variant	Arith. avg/mo	Compounded/mo	Pos. months	Sharpe	MaxDD	AnnVol
no vol target	1.65%	1.09%	74.2%	1.37	-20.2%	9.9%
60d vol estimate	1.60%	1.07%	72.0%	1.41	-15.7%	9.4%

Multiple testing, charged at the honest count: every configuration whose performance was ever computed in developing this book — the 20-cell hit-rate grid, the budget frontier, the risk-overlay, cost, execution and take-profit variants, the candidate-alpha and cross-asset studies, and the protocol re-selection — totals *109 trials, not the 8 previously reported. Recomputing the deflated Sharpe ratio on the research+validation OOS Sharpe with that count gives DSR = 0.63* (the 8-trial figure of 1.00 was flattered). A DSR near 0.6 does not certify the edge is zero; it says that after paying for the whole search, the evidence from ~4.5 years of OOS history alone leaves real uncertainty about the true Sharpe — which is why the protocol, walk-forward and bootstrap below, not a single DSR digit, carry the weight of the argument.

6.4 Three-epoch protocol: research / validation / holdout

The first edition of this report chose the sleeve budgets on a grid scored across full, in-sample and out-of-sample windows simultaneously — which means the 'OOS' numbers were used for selection and are not out-of-sample. The fix is procedural, not cosmetic. The sample is re-split into three epochs: research (2019-2022, sleeve construction and parameters), validation (2023-2024, budget selection — the only fitting on this epoch), and a holdout (2025-01 onward) scored exactly once with budgets frozen before it was touched. Re-running the same 20-cell selection with access to research + validation only picks equity 55% / crypto 30% / defensive 15% — near, but not identical, to the shipped 48/26/13 — and the frozen-budget holdout then reads:

epoch	avg month	positive months	Sharpe	max DD	months
research	1.58%	75.0%	1.23	-14.7%	48
validation	1.47%	70.8%	1.52	-11.6%	24
holdout	1.66%	81.0%	1.72	-7.9%	21
research+validation	1.54%	73.6%	1.30	-14.7%	72

The honest reading: the hit rate on research+validation is 73.6% — below the 75% bar the shipped grid cleared only by selecting on all three windows at once — while the untouched holdout came in at 81.0% positive with a 1.66% average month. Both facts are reported; the first is the price of having once searched on the later data, and the second is what a genuinely fresh window did with the same rules.

6.5 Walk-forward

A single IS/OOS cut can be lucky. The walk-forward re-runs selection as a live process would: at the start of each year, pick the budget cell on all data prior to that year (same fixed rule: hit rate first, average month as tie-break), then trade that cell through the year. Four test blocks (2023-2026) chain into one performance record:

test year	picked def/crypto	train hit rate	avg month	positive months	Sharpe	max DD
2023	15% / 30%	75.0%	1.54%	75.0%	1.43	-11.6%
2024	15% / 30%	75.0%	1.40%	66.7%	1.66	-4.5%
2025	15% / 30%	73.6%	1.84%	83.3%	1.65	-7.9%

test year	picked def/crypto	train hit rate	avg month	positive months	Sharpe	max DD
2026	15% / 30%	75.0%	1.42%	77.8%	1.89	-4.9%

Pooled across all four blocks: 75.6% positive months, 1.56% average month, Sharpe 1.61. The selection rule never moved off the same cell — evidence the frontier point is stable under expanding information sets, not a one-off sample accident — and the walk-forward book (which never saw its own test year) matches the static book's performance almost exactly.

6.6 Block bootstrap: distributions, not points

Single-number metrics inherit the luck of one sample path. A stationary block bootstrap (expected block length 20 days, 2000 resamples) over the daily net returns gives the sampling distribution of each headline metric:

metric	5th pct	median	95th pct
sharpe	0.887	1.41	1.93
max_dd	-0.227	-0.144	-0.091
pct_pos	0.624	0.699	0.774
avg_month	0.010	0.016	0.022

Three facts the distributions expose that the point estimates hide. Sharpe: the 5th percentile is 0.89 — the edge survives resampling, but a bad draw of history could plausibly have halved it. Max drawdown: the median resample draws a deeper loss (-14.4%) than the realised -15.0%, and the 5th percentile reaches -22.7% — the realised path was not the worst case. Hit rate: the 90% interval runs 62% to 77% around a median of 70% — the claimed 75% is at the optimistic edge of what this sample can support, and the honest statement is 'roughly 70%, plausibly above 75%, not demonstrably so'.

6.7 Sleeve contribution: leave-one-out

What each sleeve is actually for, tested by removing it and re-normalising the remaining budgets (holdout epoch in the last three columns, full sample for reference):

variant	avg mo	Sharpe	maxDD	pos	avg mo	Sharpe	maxDD	pos
full	1.64%	1.43	-15.0%	75.3%	1.67%	1.71	-8.5%	81.0%
minus equity core	1.98%	1.08	-19.7%	62.4%	2.51%	1.36	-11.3%	66.7%
minus btc trend	1.32%	1.32	-14.0%	72.0%	1.42%	1.71	-6.8%	76.2%
minus defensive	1.75%	1.16	-20.3%	69.9%	1.51%	1.19	-17.0%	71.4%
minus wd mom	1.57%	1.38	-14.7%	75.3%	1.66%	1.71	-7.9%	81.0%

Each sleeve earns its seat on a different axis. Removing equity raises the average month (crypto scales up into the freed gross) but cuts Sharpe from 1.43 to 1.08 and the hit rate by 13 points — equity is the stability engine. Removing defensive costs the most Sharpe in the holdout and deepens max DD from -8.5% to -17.0% — it is the crash hedge, as designed. Removing the BTC trend sleeve costs the most average month in the full sample — it is the return engine. Removing the small momentum sleeve changes almost nothing, and its retention is a diversification decision, not a return one.

6.8 Verification audit

This subsection is a self-audit against the four failure modes that matter for a backtest of this kind, with the mechanism and the evidence for each verdict. Every check below is also enforced by the automated suite in `tests/test_no_lookahead.py` (24 tests), which `make all` runs before anything else.

6.8.1 Lookahead and data leakage

The rule the engine obeys: a signal computed from the close of day t can first earn money from the open of day t+1, never earlier. Evidence, in increasing order of subtlety:

- *Signals.* Every signal function is tested for invariance: appending 60 future bars, or truncating the sample, must leave every past signal value bit-for-bit unchanged. A causal function cannot know the future, so this catches any accidental use of it.
- *Fundamentals.* The (rejected) earnings sleeve stamps each EPS value at fiscal period end plus a 60-day publication lag; a test perturbs one EPS observation and asserts the signal is untouched before the boundary and changes only after it.
- *The mixed-calendar trap.* The execution lag is counted in each asset's own trading days. A naive flat shift over a union calendar lets a Friday signal earn Friday's equity move (the crypto row on Monday is Saturday's, which forward-fills to Friday) — the regression test here planted a 5-day momentum signal on pure noise and asserts the resulting Sharpe is ~zero. Under the bug, that Sharpe was implausibly large; the fix is what ships.
- *Risk overlays.* The vol target, drawdown throttle, crash guard and budget scalers all run inside the engine, so they get their own invariance test over the final exposure panel, and the weekly W-FRI momentum formation is tested to use only completed weeks and to stay flat within the week.
- *Cash yield.* The T-bill rate published with effective date t enters the books from t+1 only, and weekends carry the last completed quote.

6.8.2 Transaction costs

Costs are charged on the bar each new position starts earning, at 1.5 bp per side on equities and 10 bp on crypto — matching or exceeding the brief's guidance — plus slippage of 2% of each asset's own daily volatility per side, which lands the all-in numbers near 4 bp (equities) and 18 bp (crypto): roughly twice the brief's floor. The cost grid confirms the reported numbers do not live off cheap-fill assumptions, and the same-close fill sensitivity in section 6.3 brackets the execution convention from the optimistic side.

6.8.3 Survivorship and delistings

The 30-coin crypto universe includes delisted Binance USDT pairs (WAVES, NEO, OMG and peers), so the cross-section a real trader faced is the one the sleeve ranks. The liquidity filter is point-in-time — trailing 20-day dollar volume only, tested by perturbing future volume — so a dying coin exits the eligible set before it stops printing. The one place survivorship remains is the equity list, which is today's large caps; the data to fix it is not free, and the direction of the bias is stated rather than bounded (section 8).

6.8.4 Overfitting

- *Parameter count is small and a priori.* EMA 50/200, RSI 14, SMA 200-day gate, 63/126/252 trend horizons, 3-week weekday momentum — literature defaults, set before the backtest, none re-estimated.

- *The sensitivity surfaces are flat where it matters* (section 6.2): equity_core's parameters can be doubled or halved with OOS Sharpe moving a few tenths. The honest exception — the crypto RSI floor — is flagged there rather than hidden.
- *IS/OOS discipline, re-established.* The first edition's budget search scored the out-of-sample windows and used them for selection; the three-epoch protocol in section 6.4 re-runs selection on research + validation only and scores 2025+ as a true frozen holdout, with a walk-forward chain (6.5) replicating the selection rule year by year.
- *Multiple testing is charged at the honest count.* 109 configurations were evaluated in developing this book; the deflated Sharpe at that count is 0.63 (section 6.3) — weaker than the 8-trial figure first reported, and stated as such. The weight of the evidence argument rests on the holdout, walk-forward and bootstrap agreement, not on one digit.
- *The hit rate is a range, not a point.* The bootstrap puts the monthly positive share at roughly 70% with a 90% interval of 62%-77% (section 6.6): the 75% headline sits at the optimistic edge of what the sample supports, and the report says so rather than quoting it to three decimals.

Verdict: the shipped book is clean on lookahead, costs and survivorship; its exposure to overfitting is disclosed and concentrated in one place — the budget selection — which is exactly why the 75.3% figure is presented as a sample property, not a strategy constant.

7. Limitations, and what a larger programme would buy

Every backtest is a compromise between the data available, the assumptions made, and the questions being asked. The limitations below are not artefacts of a rushed job; they are the boundaries of what a single-researcher, public-data project can honestly claim. Each one is stated here because it qualifies the headline result, and because a reader deserves to know precisely where the edge is soft.

7.1 Survivorship bias

The equity universe is drawn from today's large-cap constituents: AAPL, MSFT, NVDA, and the rest were all selected because they are liquid, well-known names in 2026. This flatters the equity sleeve. Companies that were large in 2019 but have since declined or been acquired are absent from the backtest universe. The fix is a point-in-time index-membership file, e.g. historical S&P 500 constituents with entry and exit dates, which is not freely available in a form suitable for this kind of causal backtest. The 2010-2016 equity-only run is reported partly to mitigate this concern: it uses the same stock list, so it inherits the same bias, and its Sharpe of 1.10 should be read as an upper bound for the equity sleeve's true historical performance.

7.2 The crypto window starts where crypto starts

The full-sample backtest begins in January 2019, which is the first date on which the Binance USDT cross-section has enough liquid names to run the momentum sleeve as designed. That is not a cosmetic choice: before 2019, the tradable crypto universe was too thin to support a weekly cross-sectional ranking. But it also means the sample starts at the beginning of a crypto bull phase. The crypto sleeve's contribution to the early years is therefore flattered by the timing. The 2017-2018 crypto market is run separately as a robustness check and is not part of the shipped result. The 2010-2016 equity-only holdout is reported for the same reason: it is the only way to audit the book outside the crypto era.

7.3 The hit rate is met by one month of margin

The positive-month count is 70 out of 93, against a requirement of 70. The condition is met by exactly one month. This thinness is reported in the summary table and here repeated: I would not represent 75% as a stable property of this strategy. On the 2010-2016 equity-only holdout, the same machine is positive in only 56% of months despite a higher Sharpe (1.10). Win rates are sample-dependent in a way that Sharpe ratios are not — and the block bootstrap in section 6.6 makes the same point quantitatively: the 90% interval for the positive-month share runs 62% to 77%. The >75% bar was cleared in the full sample; whether it clears in the next 93 months is not something any honest backtest can promise.

7.4 Fundamental data breadth

The rejected earnings sleeve used quarterly TTM EPS for 14 tickers, a fraction of the cross-section that would be needed to build a real fundamental factor. The original WorldQuant-style alpha relied on analyst estimate revisions, which are richer and more timely than realised EPS. With 14 names and realised earnings, the signal was too sparse to survive costs. This is not a failure of the idea; it is a limitation of the freely available fundamental data. A proper test would require point-in-time analyst estimates for at least the full equity universe, preferably with historical revision history.

7.5 Crypto cost assumptions are fair-weather estimates

Crypto costs are set at 10 bp per side plus slippage of 2% of daily volatility, roughly 18 bp all-in. These are reasonable for liquid large-cap pairs on a major exchange in normal conditions. They are optimistic for stressed conditions: during liquidations, spreads widen, depth evaporates, and the same fill assumption no longer holds. Crypto is the highest-turnover sleeve in the book, so the cost model is most exposed there. A conservative treatment would apply a stress multiplier to crypto costs during high-volatility regimes; that refinement is not implemented here, and the reader should treat the crypto sleeve's net returns as slightly generous during the most volatile months.

7.6 No futures, hence no cheap access to rates, commodities, and FX

The brief explicitly permits futures, and section 5.3 shows why the textbook diversifiers — rates, credit, commodities, FX — fail to help under a 100% gross cap. Their trend signals are real, but their volatilities are so low that they cannot contribute meaningful risk to a book that cannot lever. A real CTA would run 300-500% notional on these instruments; the no-leverage rule forbids exactly that. The absence of futures in the shipped book is not an oversight. It is a direct consequence of the constraint. Under a higher gross allowance, or with access to futures margin as a form of exposure rather than funding, these assets would likely improve the book's risk-adjusted return. Inside the cap as written, they are dead weight.

7.7 What a larger programme would buy

The gap between 1.64% and 2% per month is real, and it is not a tuning problem. It is a research-programme-sized gap. The following steps would be the natural next moves for a team with dedicated resources.

- *Point-in-time index membership.* Replace the current large-cap list with historical S&P 500 or Russell 1000 constituents, including delistings and corporate actions. This removes the survivorship bias in the equity sleeve and allows the backtest to run on a universe that existed at the time.
- *Execution research on crypto.* The crypto sleeve assumes 18 bp of slippage. In practice, liquid large-cap pairs on major exchanges can often be traded at 5-8 bp per side with limit orders and careful timing. Halving the slippage assumption adds roughly 0.2 percentage points to the

compounded monthly return without changing win rate or gross. This is the highest-certainty edge left on the table.
- *Defined-risk options overlays.* Options on equity indices and Bitcoin offer a way to earn volatility risk premia inside a 100% notional cap. A short-volatility or carry-style options strategy, properly sized and stress-tested, could add return without requiring additional gross exposure. This is a separate research stream, not a modification of the existing sleeves.
- *Broader data sources.* Intraday, orderbook, on-chain, and analyst-estimate data are all natural extensions. Each opens a new strategy space that daily public bars cannot reach. The current book is deliberately restricted to public daily data; a larger programme would not be.

A proper walk-forward re-estimation loop — flagged by the first edition as future work — has since been built and is reported in section 6.5. None of the remaining steps were attempted here. They are identified as the highest-value directions for future work, and the gap between the shipped result and the 2% floor is worth that programme. The result reported here is not the end of the search; it is the frontier of what could be reached with the data and constraints available to a single researcher.

Reproducibility: `make all` runs the 24 leakage tests, the backtest, the validation suite and this report from the frozen snapshot, end to end, with no network access.